

AI & ML Task 2 - House Price Prediction Model Comparison

Feature Engineering , Model Optimization & Performance
Comparison

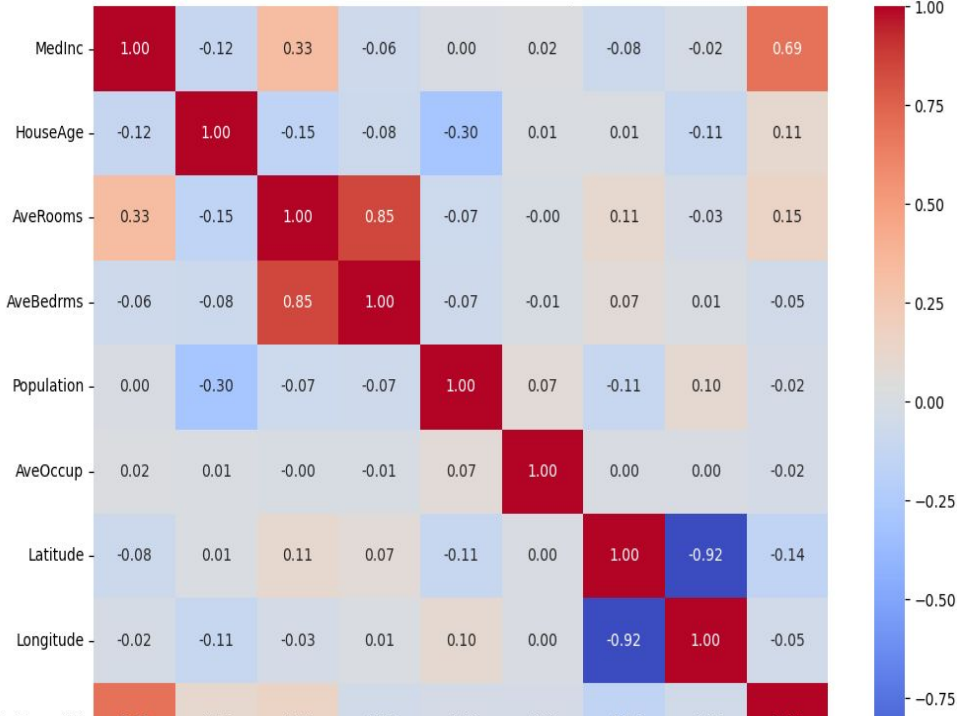
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Date:26/5/2026

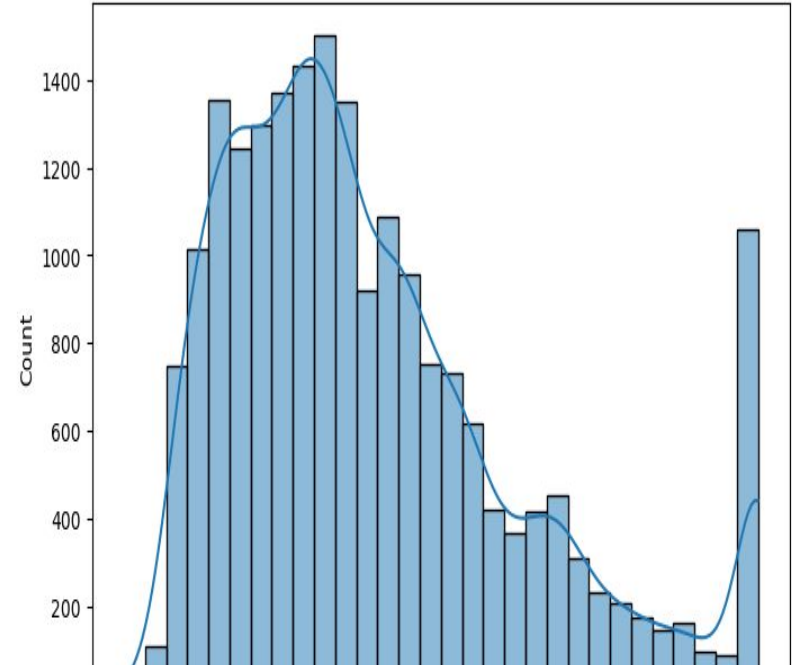
Description: "Used California Housing DataSet

"Performed EDA and applied Standard Scaling on features

Feature Correlation Heatmap



Distribution of Median House Value

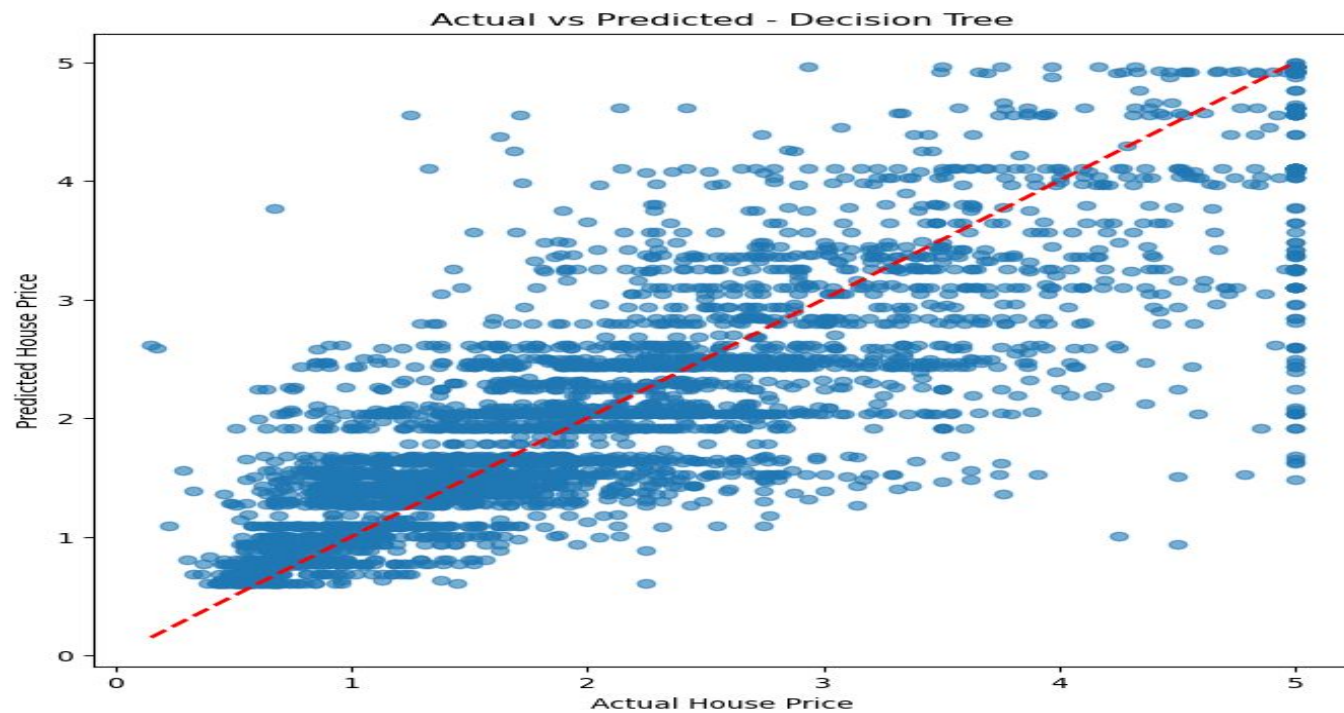


Methodology

Steps:

1. Loaded dataset using scikit-learn
2. Applied Feature Scaling using StandardScaler
3. Split data (80-20)
4. trained 3 models: Linear Regression, Ridge Regression, Decision Tree
5. "Feature scaling is important because features have different scales."

Results:



Conclusion

Best model selected based on highest R^2 score.

Key Learning: Feature scaling improves performance. Decision Tree handles non-linear data better.

Future Work: Try Random Forest, Hyperparameter tuning, more features.

"This project helped me understand real ML workflow."